

Lightning Network Economics: Channels

When Should Two Parties Use a Payment Channel?

Guasoni, Huberman, and Shikhelman (2024)

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The Bitcoin Scalability Problem

Bitcoin's Limitations:

- ~ 7 transactions/second
- High fees during congestion
- Slow confirmation times

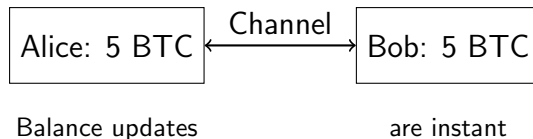
Compare to:

- Visa: $\sim 65,000$ tps
- Credit cards: instant settlement

Key Question: How can we scale Bitcoin without changing the base protocol?

Enter the Lightning Network

Core Idea: Move transactions *off-chain*



How it works:

- 1 Open channel: one on-chain transaction
- 2 Exchange unlimited payments off-chain
- 3 Close channel: one on-chain transaction

Trade-off: Save on transaction fees ↔ Lock up capital

Analogy: The Office Coffee Pool

Inefficient Approach:

- Pay \$3 each morning
- \$1 transaction fee each time
- 20 workdays = \$20 in fees

Efficient Approach:

- Prepay \$60 monthly
- \$1 fee twice (in & out)
- Save \$18 in fees
- But: lock up \$60 for a month

Lightning channels face the same trade-off

Formalizing the Problem

Two nodes exchange payments over time

Parameters:

- λ_1 : Alice pays Bob at rate λ_1 (payments/time)
- λ_2 : Bob pays Alice at rate λ_2 (payments/time)
- C : cost of one on-chain transaction
- B : cost to open/close a channel (round-trip)
- r : interest rate (opportunity cost of locked capital)
- l_1, l_2 : collateral committed by each party

Key Assumption: Payments arrive as Poisson processes (random timing)

Central Question: When is a channel cheaper than on-chain payments?

Three Ways to Pay

1 All On-Chain

- No capital locked
- Cost: $C(\lambda_1 + \lambda_2)/r$
- Simple but expensive at high volume

2 Unidirectional Channel

- Only one party makes payments through channel
- Other party pays on-chain
- Good when $\lambda_1 \gg \lambda_2$ (or vice versa)

3 Bidirectional Channel

- Both parties make payments through channel
- Benefits from *netting* opposing flows
- Best when $\lambda_1 \approx \lambda_2$

Understanding Channel Costs

A channel has two types of costs:

① Opening/Closing Cost: B

- Fixed cost to set up channel
- Amortized over channel lifetime
- Want: longer channel lifetime $\Rightarrow$ larger collateral

② Opportunity Cost: $r \times (l_1 + l_2)$

- Capital locked in channel can't be used elsewhere
- Want: smaller collateral

Trade-off:

Large collateral = long lifetime, low reset costs

Small collateral = high opportunity cost saving

Main Result: Unidirectional Channels

Setting: Alice pays Bob at rate λ , Bob never pays Alice

Theorem (Asymptotic Cost for $r \rightarrow 0$)

Optimal channel size: $l_2^* = \sqrt{B\lambda/r}$

Minimal cost: $c^* = 2\sqrt{B\lambda/r} - B/2 + O(r^{1/2})$

Key Insights:

- Cost scales as $r^{-1/2}$ (square root scaling)
- Optimal collateral also scales as $r^{-1/2}$
- As interest rate $\downarrow$, should commit more capital
- As payment rate $\uparrow$, should commit more capital

Main Result: Symmetric Bidirectional Channels

Setting: Both parties pay each other at rate λ

Theorem (Asymptotic Cost for $r \rightarrow 0$)

Optimal channel sizes: $l_1^* = l_2^* = (2B\lambda/r)^{1/3} + O(r^{1/3})$

Minimal cost: $c^* = 3(2B\lambda/r)^{1/3} - B/6 + O(r^{1/3})$

Key Insights:

- Cost scales as $r^{-1/3}$ (cubic root scaling)
- **Much better than unidirectional!** ($-1/3$ vs $-1/2$)
- Benefit from *netting*: opposing flows cancel out
- Both parties commit capital

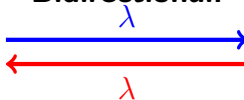
Why Does Bidirectional Scale Better?

Unidirectional:



Balance depletes at rate λ

Bidirectional:



Net flow = 0
Balance depletes slower!

Mathematical reason:

- Unidirectional: Balance follows one-sided random walk
- Bidirectional: Balance follows two-sided random walk
- Takes longer to hit boundary $\Rightarrow$ longer lifetime $\Rightarrow$ lower cost

Asymmetric Bidirectional Channels

Setting: $\lambda_2 > \lambda_1 > 0$ (unequal payment rates)

Theorem (Asymptotic Cost for $r \rightarrow 0$)

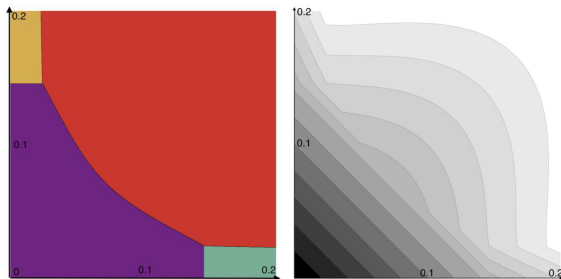
If $l_2 = O(r^{-1/2})$ and $l_1 = O(\log(r^{-1}))$, then:

$$\text{Minimal cost: } c^* = 2\sqrt{B(\lambda_2 - \lambda_1)/r} + \frac{1 + \log(1 + \sqrt{B(\lambda_2 - \lambda_1)/r \cdot \log(\lambda_2/\lambda_1)})}{\log(\lambda_2/\lambda_1)} + O(1)$$

Key Insights:

- Cost scales as $\boxed{r^{-1/2}}$ (like unidirectional!)
- Asymmetric channels behave like unidirectional with flow $\lambda_2 - \lambda_1$
- Average payer commits $O(r^{-1/2})$ capital
- Average payee commits only $O(\log(r^{-1}))$ capital
- Netting benefit exists but doesn't change scaling

Synthesis: When to Use Which Channel?



Four regions in (λ_1, λ_2) space:

- 1 **Bottom-left:** Both rates low $\Rightarrow$ All on-chain
- 2 **Top-left/Bottom-right:** One rate high $\Rightarrow$ Unidirectional
- 3 **Top-right:** Both rates high $\Rightarrow$ Bidirectional
- 4 **Transition zones:** Hybrid strategies

When Are Channels Worth It?

Necessary conditions for channel optimality:

Theorem (Lower Bounds on Payment Rates)

- ① Unidirectional channel: $C > \frac{Br}{\lambda} + o(r)$
- ② Bidirectional symmetric: $C > 3\left(\frac{Br^2}{4\lambda^2}\right)^{\frac{1}{3}} - \frac{Br}{12\lambda} + o(r)$

Interpretation:

- Channels only worth it when:
 - On-chain costs (C) are high, or
 - Channel reset costs (B) are low, or
 - Payment rates (λ) are high, or
 - Interest rates (r) are low

Impact on Blockchain Congestion

Key Question: How many on-chain transactions do channels save?

Proposition (Congestion Reduction)

Long-run ratio of channel transactions to on-chain transactions:

- Unidirectional: l_1 (linear in collateral)
- Symmetric bidirectional: $l_1 l_2$ (product of collaterals!)

Example: $l_1 = l_2 = 10$

- Unidirectional: 10 channel txs per on-chain tx
- Bidirectional: 100 channel txs per on-chain tx

Implication: Symmetric bidirectional channels provide *quadratic* reduction in blockchain load!

Technical Contribution: The Model

This paper adapts classic inventory models to cryptocurrencies:
Baumol-Tobin (1952-56):

- Demand for cash
- Transaction costs vs opportunity cost
- Inspired unidirectional analysis

Miller-Orr (1966):

- Firm cash management
- Two-sided random walk
- Inspired bidirectional analysis

Key differences:

- Can always transact on-chain (alternative exists)
- Both cost and value storage
- Network effects matter
- Dynamic topology

Mathematical tool: Difference equations with boundary conditions

What's Really New Here?

Beyond adapting classic models:

- ① **Scaling laws:** $r^{-1/2}$ vs $r^{-1/3}$ characterize fundamental difference
 - Not obvious from Baumol-Tobin or Miller-Orr
 - Shows power of symmetry/netting
- ② **Asymmetric analysis:** Logarithmic collateral for payee
 - Surprising result: payee commits almost nothing
 - Shows channels can be very asymmetric
- ③ **Necessary conditions:**
 - Quantifies when channels are viable
 - Guides network design
- ④ **Congestion reduction:** Quadratic for symmetric bidirectional
 - Shows dramatic scalability improvement
 - Justifies Lightning Network development

What's Missing?

Model limitations:

- Unit transaction size (simplified)
- Fixed costs B, C (actually depend on congestion)
- Two-party channels (network effects not modeled)
- Poisson arrivals (may not capture all patterns)
- No routing fees
- No strategic behavior

Companion paper (Guasoni, Huberman, Shikhelman 2023):

- Analyzes Lightning Network *topology*
- How should channels connect?
- Optimal network structures

Testing the Theory

What would we observe if this model is right?

- 1 Channel sizes should scale as $r^{-1/2}$ (unidirectional) or $r^{-1/3}$ (bidirectional)
- 2 Symmetric channels should be larger and last longer
- 3 Channels should be rare when payment rates are low
- 4 Average payees should commit less than average payers

Challenges for empirical testing:

- Can't observe payment rates directly (privacy!)
- Channel sizes are public but balances are not
- Interest rate r unclear (opportunity cost subjective)
- Selection effects: who uses Lightning?

Discussion Questions

- 1 The model assumes payment rates are *exogenous*. But wouldn't the existence of cheap channels *increase* payment rates? How would endogenizing λ change the results?
- 2 The paper treats B and C as fixed. But both depend on blockchain congestion, which channels reduce. Can the feedback loop be modeled?
- 3 In the asymmetric case, the payee commits only $O(\log r^{-1})$ capital. Is this realistic? What might prevent such asymmetric arrangements?
- 4 The paper shows bidirectional channels reduce congestion quadratically. But if everyone uses bidirectional channels, what happens to the *topology* of the network?

Beyond Bitcoin

This analysis applies to any layer-2 solution:

- Ethereum: Raiden Network, state channels
- Other cryptocurrencies: Liquid Network (Bitcoin), Polygon (Ethereum)
- Even traditional finance: settlement batching, clearinghouses

General principle:

When you have:

- High per-transaction costs
- Frequent bilateral transactions
- Ability to cryptographically secure off-chain state

Then you should consider a channel-like solution

This paper quantifies exactly when and how much you save

Summary

Key takeaways:

- 1 Lightning channels trade off transaction costs for opportunity costs
- 2 Unidirectional channels: cost $\sim r^{-1/2}$, good for one-way flows
- 3 Bidirectional channels: cost $\sim r^{-1/3}$, much better for symmetric flows
- 4 Asymmetric channels behave like unidirectional
- 5 Channels viable when payment rates high, costs high, or interest rates low
- 6 Symmetric bidirectional channels provide quadratic congestion reduction

Broader contribution:

- Rigorous economic foundation for Lightning Network
- Adapts classic inventory models to crypto context
- Provides design principles for layer-2 solutions